

Does Time Travel Imply NP is in BQP?

or

How I learned to stop worrying about patricidal paradoxes and love closed timelike curves.

Dave Bacon

Institute for Quantum Information, Caltech

Preskill Group meeting, September 13, 2001

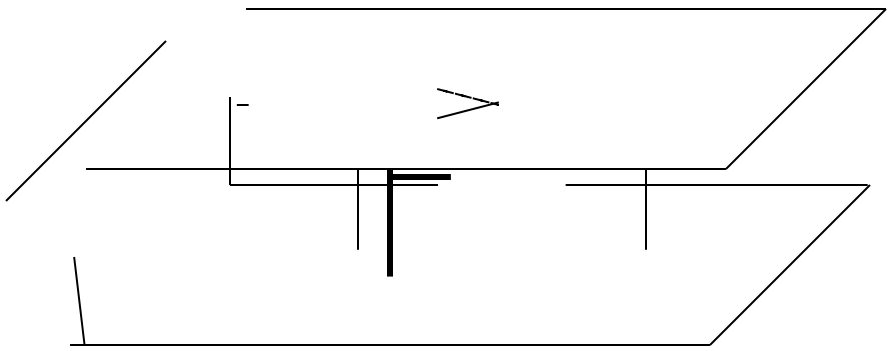
Seminar on time travel to be held three weeks ago.

VOLUME 61, NUMBER 13 PHYSICAL REVIEW LETTERS 26 SEPTEMBER 1988

Wormholes, Time Machines, and the Weak Energy Condition

Michael S. Morris, Kip S. Thorne, and Ulvi Yurtsever
Theoretical Astrophysics, California Institute of Technology, Pasadena, California 91125
(Received 21 June 1988)

*Normally theoretical physicists ask, "What are the laws of physics?" and/or "What do those laws predict about the Universe?" ... We ask, instead, "**What constraints do the laws of physics place on the activities of an arbitrarily advanced civilization?**" ... whether the laws of physics permit an arbitrarily advanced civilization to construct and maintain wormholes.* Morris, Thorne, Yurtsever PRL, **61**, 1446 (1988)



Argue: stable wormholes can exist, thread with averaged weak energy condition violating material:

$$\int T_{\alpha\beta} k^{\alpha} k^{\beta} d\varsigma < 0, \quad k^{\alpha} = \frac{dx^{\alpha}}{d\varsigma}$$

Gertrude Stein



"There are two theres there."

Turning Wormholes into Time Machines

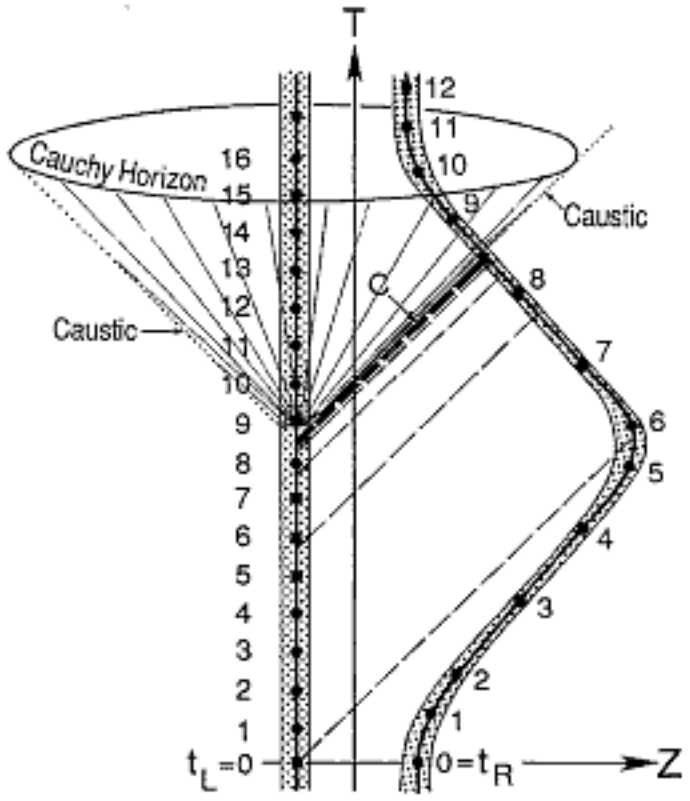


FIG. 2. Spacetime diagram for conversion of a wormhole into a time machine.

Morris, Thorne, Yurtsever (1984)

win paradox"
could use gravitational redshift)

elay between wormhole throats

conditions uniquely determine
on up to "Cauchy Horizon"

y Horizon is set of measure
MTY conjecture unique
ion through the Cauchy
Horizon.

Horizon is stable? …

I am **NOT** the Mighty Stephen Hawking

(http://www.mchawking.com/)





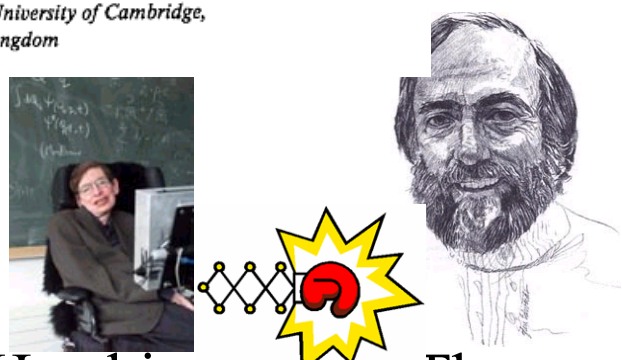
PHYSICAL REVIEW D VOLUME 46, NUMBER 2 15 JULY 1992

Chronology protection conjecture

S. W. Hawking
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Silver Street, Cambridge CB3 9EW, United Kingdom
(Received 23 September 1991)

Quantum vacuum polarizations prevent formation of closed timelike curves.

General result, no CTCs, still conjecture, no proof (yet).

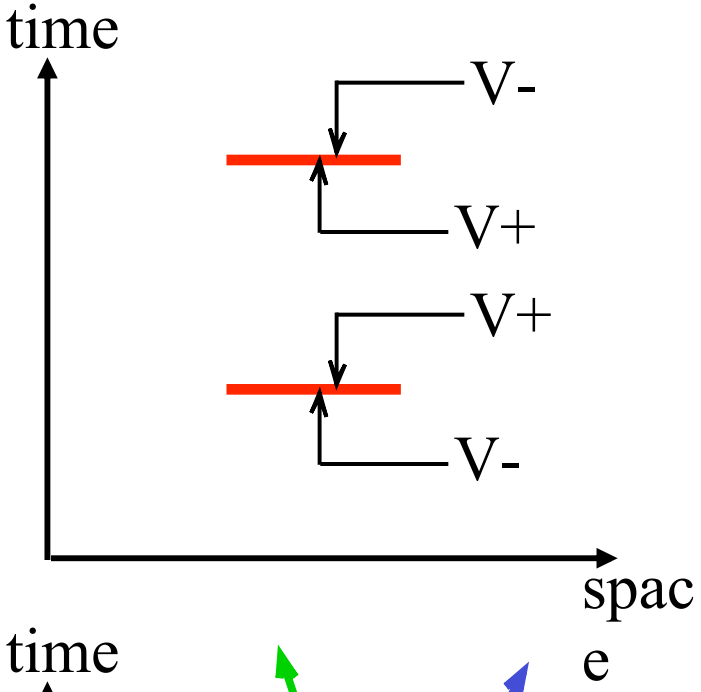


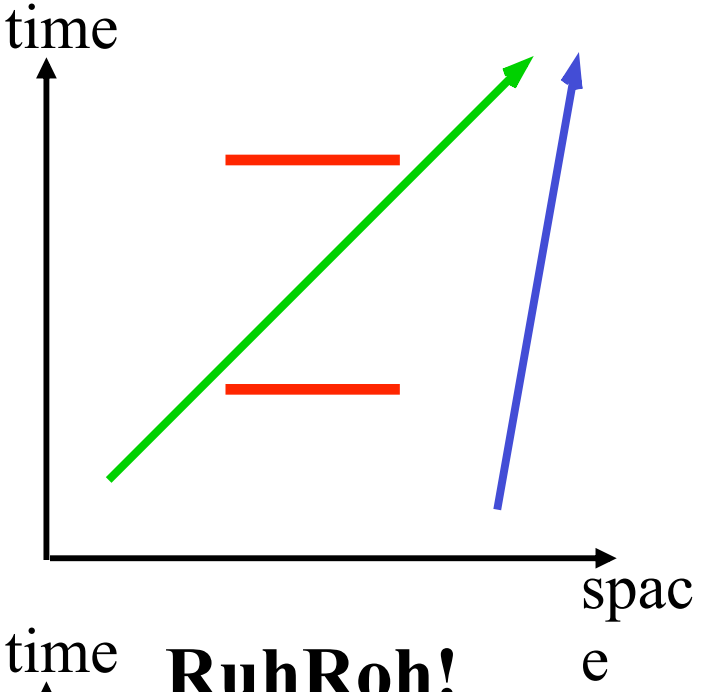
Hawking Thorne

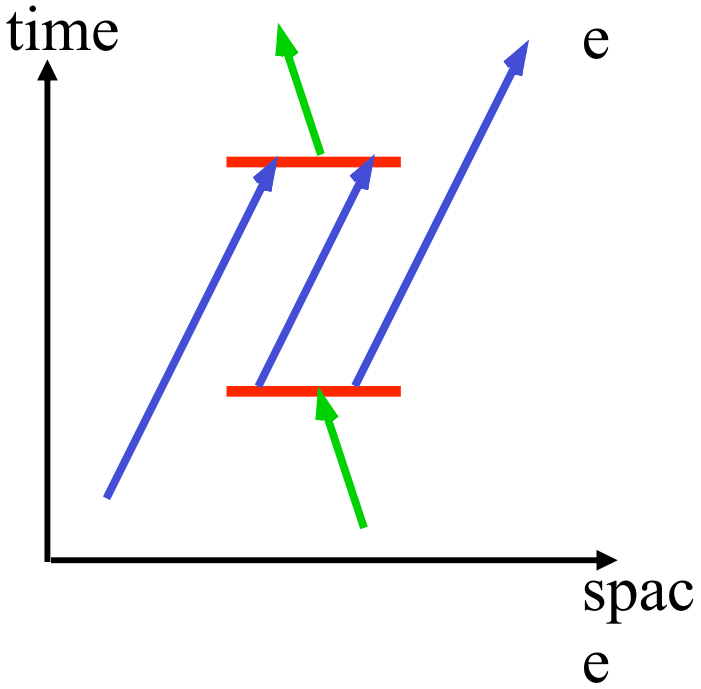
FOX 11 LOS ANGELES "Tonight on FOX:"

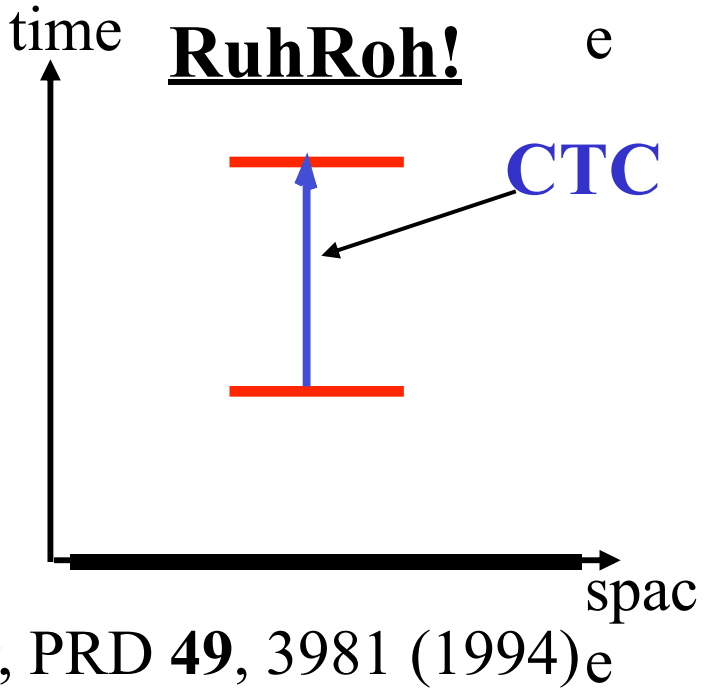
When Scientists

Closed Timelike Curves



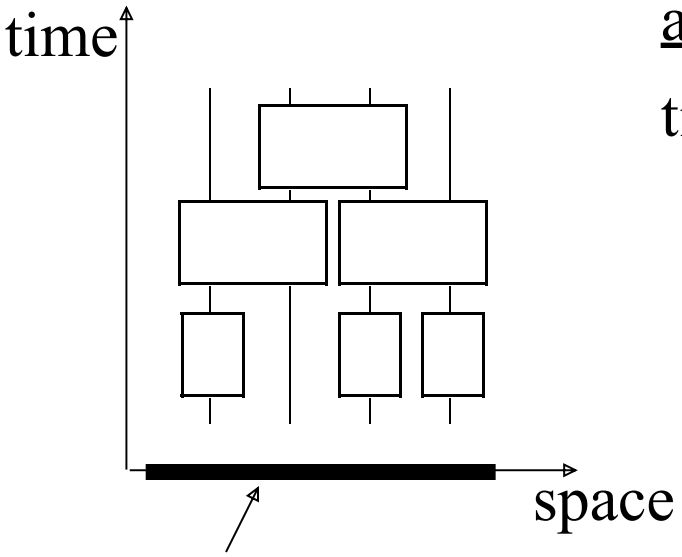






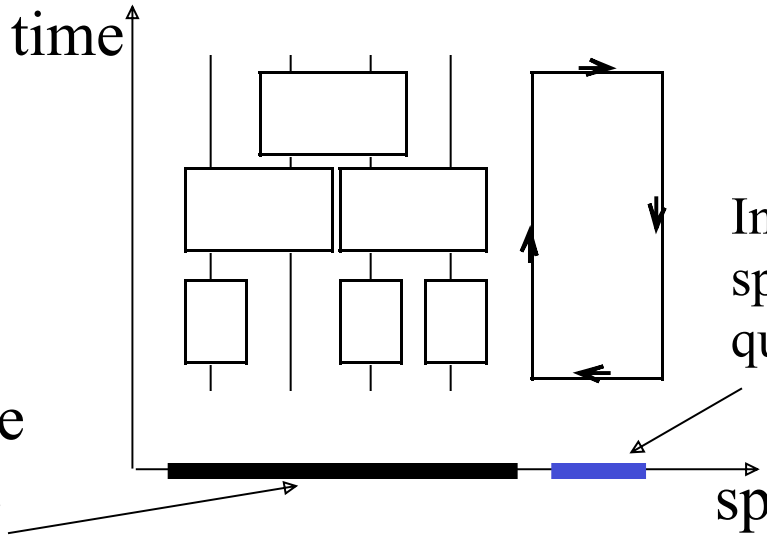
Politzer, PRD **49**, 3981 (1994)

Qubits in Spacetime



Initial conditions specifying gates and input.

allow qubits which travel on CTCs

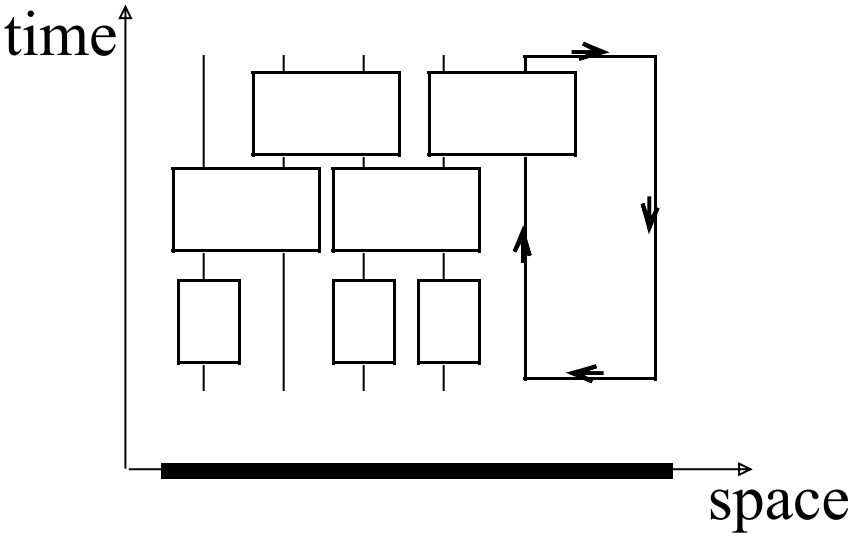


Initial conditions specifying CTC qubit (input?!)

allow CTC qubits to interact with nonCTC qubits

Complexity: resource quantification as size of computational problem

Argue: some fixed energy expendature to create CTC qubit



Consistency

To | Quantum Theory | **add** | Consistency | in the presence of | Closed Timelike Curves |

Principle of self-consistency

"the only solutions to the laws of physics that can occur locally are those which are globally self-consistent."

I.D. Novikov, *Evolution of the Universe* (1983)

Friedman *et al.* "Cauchy problem in spacetimes with closed timelike curves" PRD **42** 1915 (1990)

D. Deutsch "Quantum mechanics near closed timelike lines" PRD **44** 3197 (1991)

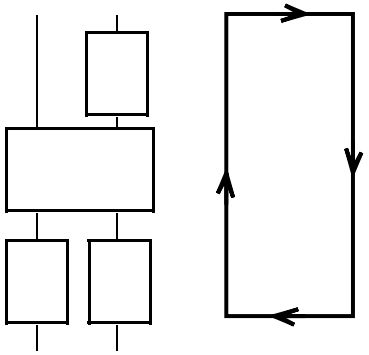


"A foolish consistency is the hobgoblin of little minds, adored by little statesmen and philosophers and diviners."

Emerson

EZ Consistency 123

Case I (BORING)

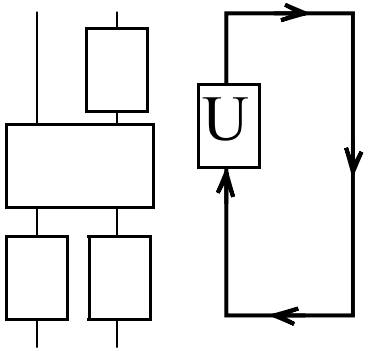


No operations on the CTC qubit

All CTC qubit states are consistent.

Output of circuit unaffected by existence of CTC

Case II (LESS BORING)



Operation on the CTC qubit

Any mixture of eigenstates of U consistent (if degeneracy, then superpositions of eigenstates)

$$\mathbf{U}\rho\mathbf{U}^{\dagger} = \rho$$

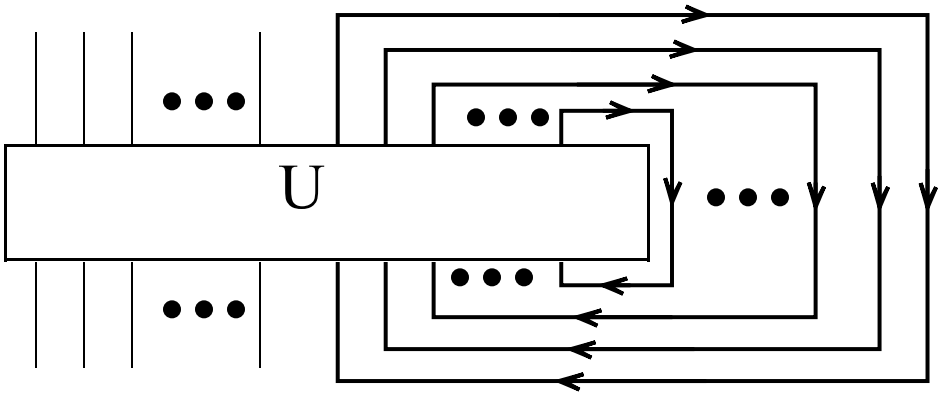
Output of circuit unaffected by existence of CTC

Question of temporal origin? $\mathbf{U} = \mathbf{U}_2\mathbf{U}_1$ or $\mathbf{U} = \mathbf{U}_1\mathbf{U}_2$

"no unique time ordering of spacelike surfaces"

Case III (OHSO NONBORING)

ρ_{out}



Most general circuit

ρ_{in} ρ

$H_A \otimes H_B$

Consistency: $\rho = \mathrm{Tr}_A \left[\mathbf{U} \left(\rho_{in} \otimes \rho \right) \mathbf{U}^\dagger \right]$

Output: $\rho_{out} = \mathrm{Tr}_B \left[\mathbf{U} \left(\rho_{in} \otimes \rho \right) \mathbf{U}^\dagger \right]$

Question: Is there always a consistent solution?

Question: More than one consistent solution?

"42," said Douglas Adams

D. Deutsch "Quantum mechanics near closed timelike lines" PRD **44** 3197 (1991)

Fixed point of superoperator *S*:

$$S[\rho] = \mathrm{Tr}_A\left[\mathbf{U}\left(\rho_{in}\otimes\rho\right)\mathbf{U}^{\dagger}\right]$$

Define density operator $\rho(N)$ for arbitrary$\rho(0)$:

$$\rho\left(N\right) = \frac{1}{N+1}\sum_{n=0}^{N} S^n\left[\rho\left(0\right)\right]$$

Sequence $\{\rho(N)\}$ must have at least one accumulation point ρ_{lim}

Bound:

$$0 \le \mathrm{Tr}\left[\left(S\left[\rho\left(N\right)\right]-\rho\left(N\right)\right)^2\right]$$

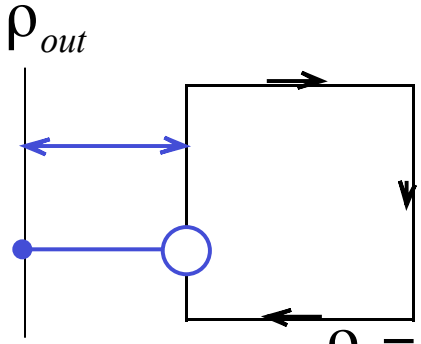
$$= \frac{1}{N+1}\mathrm{Tr}\left[\left(S^{N+1}\left[\rho\left(0\right)\right]-\rho\left(0\right)\right)^2\right] \le \frac{2}{N+1}$$

Such that:

$$\mathrm{Tr}\left[\left(S\left[\rho_{\mathrm{lim}}\right]-\rho_{\mathrm{lim}}\right)^2\right] = 0 \rightarrow S\left[\rho_{\mathrm{lim}}\right] = \rho_{\mathrm{lim}}$$

Argument by Example

two qubits



$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Consistency implies:

$$m_x = n_x m_z \quad m_y = n_y m_z \quad m_z = n_z$$

single consistent CTC qubit states.

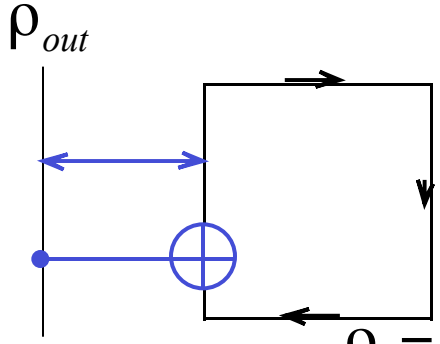
Output is unambiguous though:

$$\rho_{out} = \frac{1}{2}\left(\mathbf{I} + {n_z}^2 n_x \sigma_x + {n_z}^2 n_y \sigma_y + n_z \sigma_z\right)$$

NONLINEAR transformation:

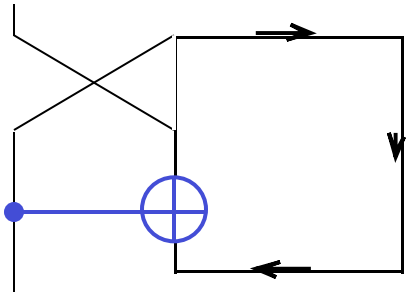
$$\frac{1}{2}(\mathbf{I} + n \cdot \sigma) \rightarrow \frac{1}{2}\left(\mathbf{I} + {n_z}^2 n_x \sigma_x + {n_z}^2 n_y \sigma_y + n_z \sigma_z\right)$$

"Controlled-Not"ing Oneself



$$\rho_{in} = \tfrac{1}{2}\left(\mathbf{I} + n \cdot \sigma\right)$$

$$\rho = \tfrac{1}{2}\left(\mathbf{I} + m \cdot \sigma\right)$$



$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Consistency implies:

$$m_x = n_x m_x \quad m_y = n_y m_x \quad m_z = n_z$$

Output is unambiguous **if** $nx \neq 1$

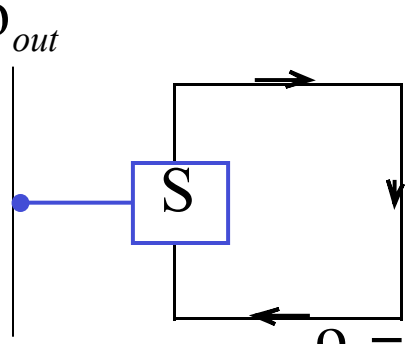
$$\rho_{out} = \tfrac{1}{2}\left(\mathbf{I} + n_z{}^2 \sigma_z\right)$$

however CTC qubit and output is ambiguous **if** $nx = 1$

$$\rho_{out} = \tfrac{1}{2}\left(\mathbf{I} + m_x \sigma_x\right) \qquad \rho = \tfrac{1}{2}\left(\mathbf{I} + m_x \sigma_x\right)$$

Question: What to do with ambiguous solutions?

“Controlled-S”ing



$$\rho_{in} = \tfrac{1}{2}\left(\mathbf{I} + n \cdot \sigma\right)$$

$$\rho = \tfrac{1}{2}\left(\mathbf{I} + m \cdot \sigma\right)$$

$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & I \end{bmatrix}$$

$$m_x = m_x\left(\frac{1+n_z}{2}\right) + m_y\left(\frac{-1+n_z}{2}\right) \qquad m_y = m_x\left(\frac{1-n_z}{2}\right) + m_y\left(\frac{1+n_z}{2}\right) \qquad m_z = m_z$$

$$\rho_{out} = \tfrac{1}{2}\left(\mathbf{I} + \left[\left(\frac{n_x - n_y}{2}\right) + m_z\left(\frac{n_x + n_y}{2}\right)\right]\sigma_x + \left[\left(\frac{n_x + n_y}{2}\right) + m_z\left(\frac{-n_x + n_y}{2}\right)\right]\sigma_y + n_z\sigma_z\right)$$

Ambiguity not just for a single input state.

Ambiguous for all input states except $nx = ny = 0$

Deutsch's Ambiguous Solution?



David Deutsch

"[Philosophical posture deleted]…*the state of the supplementary data is the state of greatest entropy compatible with the initial data*."

$$\text{i.e.} \max_{\rho|\rho \text{ is consistent}} \left(-Tr\rho \ln \rho\right) \quad (1)$$

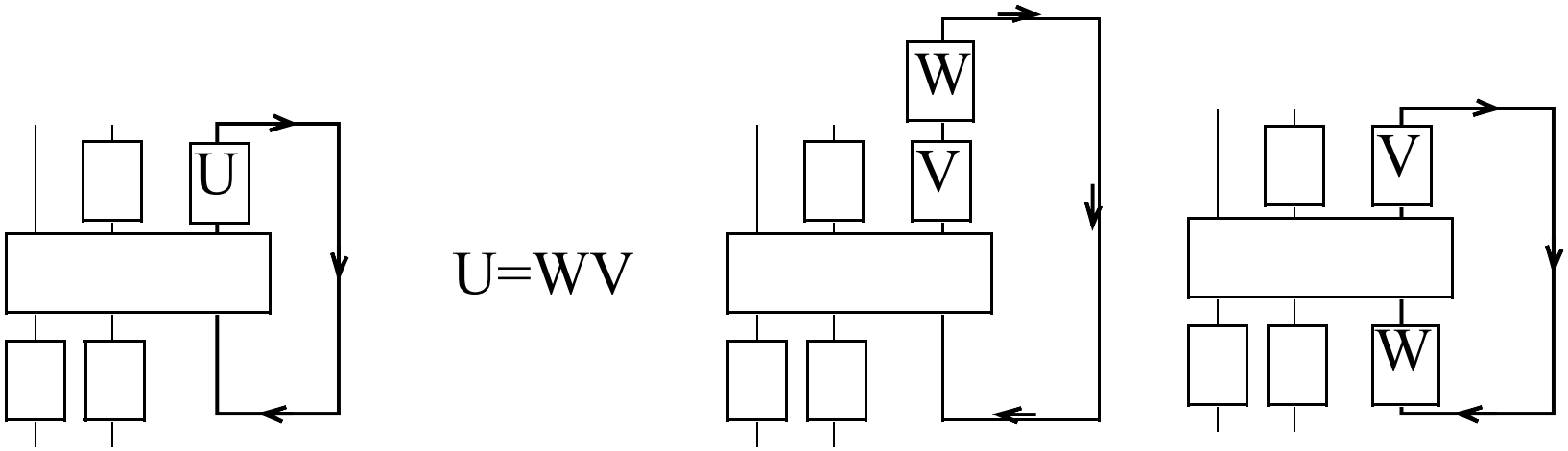
Open Question I: Is Deutsch's solution to the consistency condition ambiguous? Are there multiple solutions to (1) which lead to different $\rho_{in} \square \rho_{out}$ mappings?

Open Question II: Can the ambiguity be explained via evolution from initial conditions which created the CTC qubits?

See Politzer, PRD **49**, 3981 (1994) for another view on the ambiguity

Dave, look at your watch to see how long you've been yacking.

What Time is it?



Consistency:

$$\rho = \mathrm{Tr}_A\left[(\mathbf{I}\otimes\mathbf{U})\left[\overline{\mathbf{U}}(\rho_{in}\otimes\rho)\overline{\mathbf{U}}^{\dagger}\right](\mathbf{I}\otimes\mathbf{U})^{\dagger}\right]$$

$$\rho = \mathrm{Tr}_A\left[(\mathbf{I}\otimes\mathbf{WV})\left[\overline{\mathbf{U}}(\rho_{in}\otimes\rho)\overline{\mathbf{U}}^{\dagger}\right](\mathbf{I}\otimes\mathbf{WV})^{\dagger}\right]$$

$$\rho = \mathrm{Tr}_A\left[(\mathbf{I}\otimes\mathbf{V})\left[\overline{\mathbf{U}}\left[(\mathbf{I}\otimes\mathbf{W})(\rho_{in}\otimes\rho)(\mathbf{I}\otimes\mathbf{W})^{\dagger}\right]\overline{\mathbf{U}}^{\dagger}\right](\mathbf{I}\otimes\mathbf{V})^{\dagger}\right]$$

Open Question III: Temporal origin problem?

Nonlinear Evolution

$$|0\rangle \rightarrow |1\rangle$$
$$|1\rangle \rightarrow |0\rangle$$

BUT

$$\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \rightarrow |0\rangle$$

(mostly discussed in context of "solution to measurement problem")

"If you hit a tuning fork twice as hard it will ring twice as loud. That's a linear response. If you hit a person twice as hard, they're unlikely just to shout twice as loud. That property lets you learn more about the person than the tuning fork" - Neil Gershenfeld.

“Superman don’t need no seat belt”-Muhammad Ali

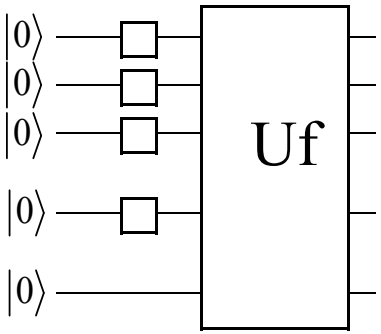
Nonlinear evolution of a quantum system *can* lead to efficient solution of NP-complete and #P-complete problems.

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Nonlinear Quantum Mechanics Implies Polynomial-Time Solution for *NP*-Complete and #*P* Problems

Daniel S. Abrams*
Department of Physics, MIT 12-128b, Cambridge, Massachusetts 02139

Seth Lloyd†
d’Arbeloff Laboratory for Information Sciences and Technology, Department of Mechanical Engineering, MIT 3-160, Cambridge, Massachusetts 02139
(Received 27 January 1998)



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|f(x)\rangle$$

The Superman Nonlinear Transform

$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|0\rangle \rightarrow |0\rangle|0\rangle$$

$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|f(x)\rangle \rightarrow |0\rangle|1\rangle \quad \text{when } \exists x \mid f(x)=1$$

Just a Spoonful of Nonlinearity Makes the Medicine Go Down

Abrams and Lloyd showed a single qubit nonlinear transformation could be used to solve NP-complete problems:

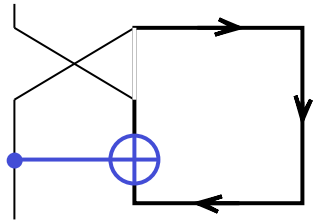
$$|0\rangle \rightarrow |0\rangle$$
$$|1\rangle \rightarrow |0\rangle$$

BUT

$$\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \rightarrow |1\rangle$$
$$\frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right) \rightarrow |1\rangle$$

Specific nonlinear QM (Weinberg, Ann. Of Phys. **194** 336 (1989)) can produce needed transformations.

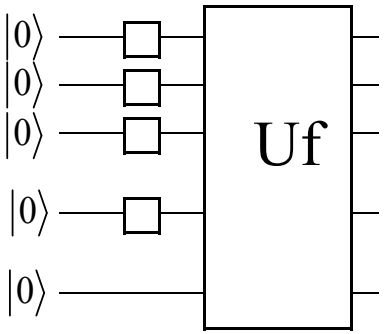
got nonlinearity?



$$\tfrac{1}{2}\left(\mathbf{I} + n\cdot\sigma\right) \xrightarrow{T} \tfrac{1}{2}\left(\mathbf{I} + {n_z}^2\sigma_z\right) \quad n_x \neq 1$$

---

n



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}\left|x\right\rangle\left|f\left(x\right)\right\rangle$$

$$\rho_0 = \begin{bmatrix} 1-\dfrac{s}{2^n} & 0 \\ 0 & \dfrac{s}{2^n} \end{bmatrix}$$

s is number of x's such that *f(x)*=1

$$\rho_0 = \frac{1}{2}\left[\mathbf{I} + \left(1 - \frac{s}{2^{n-1}}\right)\sigma_z\right]$$

got nonlinearity?

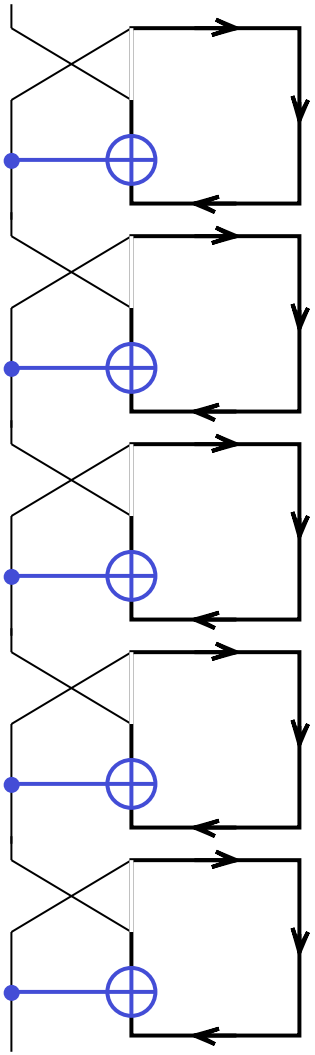
Apply T k times:

$$T^k\left[\rho_0\right] = \frac{1}{2}\left[\mathbf{I} + \left(1 - \frac{s}{2^{n-1}}\right)^{2^k} \sigma_z\right]$$

If s=0, state is still $|0\rangle$. If s ≠0, state is exponentially close, for polynomial k, to 1/2($|0\rangle\langle 0|+|1\rangle\langle 1|$).

Polynomial application of routine will determine whether s=0.

$$PROB(\text{false postive}) = \frac{1}{2^r}\left[1 + \left(1 - \frac{s}{2^{n-1}}\right)^{2^k}\right]^r$$



Time Machines imply NP is in BQP

$NP \subseteq BQPTT????$

"For a successful technology, reality must take precedence over public relations, for nature cannot be fooled." - Richard P. Feynman

Collaborator: Dr. Dr. Julia Kempe, U.C. Berkeley



CTC Guru: Mike (Jun) Cai a.k.a. STUDBUD